

Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph

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ABSTRACT

We present a new approach using Virtual Reality to capture human-scene interaction data within 3D environments. Traditional methods required costly physical setups with real furniture, making it difficult to collect both interaction and 3D scene data comprehensively. Our solution implements a user-centric scene graph with a hierarchical structure that captures both user interactions and 3D scene elements. The resulting dataset contains real human interaction data, dynamic scene information, synthetic sensor data, and scene graph annotations. This work provides a foundation for developing Mixed Reality experiences using 3D scene graphs with human interaction data.

Index Terms: Human-Scene Interaction, Virtual Reality, Augmented Reality, Scene Graph, Dataset.

1 INTRODUCTION

In Mixed Reality (MR) devices, comprehensive understanding of the user's environment is crucial, requiring both scene information and user interaction data. While human-scene interaction data is essential for system understanding, obtaining detailed real-world data is costly and time-consuming. We introduce a Virtual Reality (VR)-based data generation pipeline to create such datasets cost-effectively. Although previous human-scene interaction datasets [1] exist, they primarily focus on 3D body reconstruction or motion synthesis. Existing datasets lack detailed annotations of both spatial scene and human interaction data. Current methods require complex real-world configurations and hardware setups [1], with particular challenges in tracking object poses and 3D reconstruction. We utilize a VR system with a game engine, requiring only a head-mounted display (HMD) and controllers. We collected 5.9 hours of human-scene interaction data from 22 participants across 11 scenes, involving 36 objects, 15 interactions, and 11 spatial relationships. This generated 232 activity datasets, with 77.2% featuring dynamic scenes. Our main contributions are threefold. First, a low-cost human-scene interaction dataset generation pipeline. Second, a user-centric scene graph for efficient annotation, and third, a collection of actual human-scene data with the proposed pipeline. This approach enables cost-effective collection of comprehensive interaction and scene data, supporting natural interactions and adaptive interfaces in MR devices.

2 RELATED WORK

2.1 Human-Scene Interaction Dataset

Building models that work in general situations requires large and diverse datasets. While previous datasets[1] have collected from

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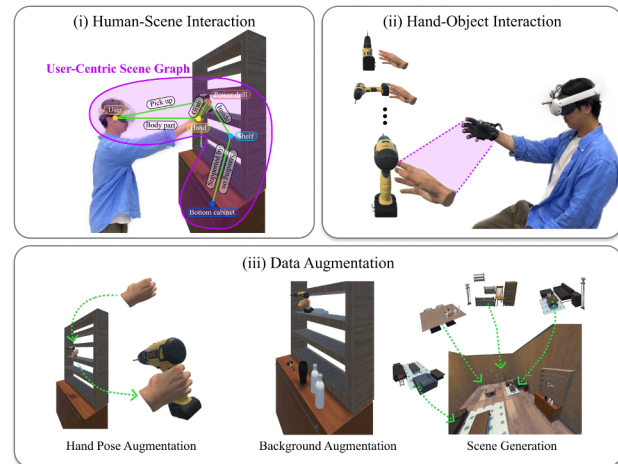


Figure 1: **The overview of sequential human-scene interaction data generation pipeline.** (i) Human-scene interaction data were collected using VR HMDs and controllers. (ii) Hand-object interaction collection. (iii) Data augmentation is performed in three ways: hand pose augmentation, background augmentation, and scene generation.

real-world activities, they cannot cover all new scenarios or environments. When researchers need data for new environments, they must build new datasets, which is costly due to physical setup and tracking requirements. To address these limitations, synthetic data methods have emerged. Lin et al. [3] proposed an VR framework for collecting real-human interaction in synthetic scenes, though it had limited scope. Our approach expands on this concept with broader scene coverage and annotation diversity.

2.2 Scene Graph

The scene graph structure has proven effective in SLAM fields for handling large-scale data, enabling accurate and efficient representations. Scene graphs are also valuable for representing human-scene interactions [2], as their edge effectively capture relationships between users and objects. While scene graphs can effectively represent both scene and human data, their application in MR devices remains underexplored. Since MR devices require understanding scenes from the user's perspective, we introduce a user-centric scene graph structure that represents spatial and interaction data with the user at the center.

3 METHOD

3.1 Data Collection

The data collection involved two processes: human-scene interaction collecting and hand-object collecting as shown in Fig. 1. The human-scene interaction process captured user movement, while the hand-object process captured hand-object interaction data. This approach allowed detailed hand pose collection without restricting

user movement in the scene. Human-scene interaction collection used cluster-scale scenes, which are smaller than room-scale. Template scenes were prepared with predefined furniture arrangements and manually annotated spatial relationships. Subjects wore VR HMDs and used controllers to trigger virtual hand interactions. Interactions were recorded at two levels: atomic actions and activities. For hand-object interaction collection, subjects wearing VR HMDs were instructed to grab virtual objects from various angles. Hand tracking was used to manipulate virtual hands, with subjects grasping objects as if real. Grab poses were recorded at the moment of subject response, with objects rotated to capture various angles. Data was collected for one hand and mirrored for the other side

3.2 User-Centric Scene Graph

During data collection, all data and annotations are stored in a user-centric scene graph structure. One of characteristics of a user-centric scene graph is that the user node is a root node for the graph. In conventional 3D scene graph, the scene node becomes the root node, and the user node becomes one of the leaf nodes. Data from MR devices is mainly obtained from the egocentric view, and the data should be calculated to transform it into world coordinates in the conventional scene graph. The user-centric scene graph reduces the transformation process and provides geometric information that can be used directly from the user's perspective. Another characteristic of a user-centric scene graph is a new hierarchy called a cluster. Cluster nodes contain children that form a connected graph. Since users typically interact with objects in their immediate surroundings, rather than updating information about every object in a room, the cluster concept efficiently partitions space and objects at a sub-room level. This structure allows identification of interactable clusters based on user location and selective updating of only relevant object information.

4 HUMAN-ENVIRONMENT INTERACTION DATASET

To validate the proposed user-centric scene graph structure and data generation pipeline, we collected a human-scene interaction dataset from recruited subjects, capturing real human-scene interaction in a user-centric scene graph format. Virtual scenes were constructed using the iGibson [5] dataset, with scenes extracted at cluster-scale. We used eight living room clusters and three kitchen clusters. The scenes comprised 23 types of furniture objects and 13 placed objects, including tools for various interactions. There were 11 classes of relationships between user and object. Atomic actions were represented by four edge types between hands and objects (grab, release, push, pull), while 11 classes of spatial relationships existed between objects. The dataset contained 27 total edge label classes including edge between user and hand('body part'). Data collection involved 22 subjects (9 males, 13 females). Collection used Meta Quest 3 VR HMD and controllers for human-scene interactions, and MANUS Prime3 Haptic XR gloves with Optitrack Motion Capture System for hand-object interactions. Each subject performed actions in 11 scenes, with five to six action sequences taking approximately five seconds each. For hand-object interaction, subjects performed grab poses on 10 objects.

5 RESULTS

Through data collecting, we annotated a total of 5.9 hours of user-scene interaction data, comprising 2,972 atomic action clips with an average of 512 frames per clip. The dataset contains 232 activities, each averaging five action sequences. In the hand-object interaction collecting, an average of 113 grab hand poses per object was collected. To evaluate the dataset, we implemented a scene graph prediction model based on [4]. The model achieved a top-1 recall ($R@1$) of 92.94% for relationship prediction using an 80-20 train-test split.

6 DISCUSSION

Our data collection pipeline generated a comprehensive human-scene interaction dataset using a user-centric scene graph format, which enables various types of annotations that extend existing datasets. Scene graph prediction evaluation achieved an $R@1$ value of 92.94%, demonstrating our dataset's effectiveness for training prediction models. The evaluation showed that our dataset, which includes both interaction and spatial relationships, enables comprehensive relationship prediction. This validates that user-centric scene graphs can serve as effective data representations for human-scene interactions.

Our study has two main limitations. First, it lacks a baseline model or benchmark for performance evaluation. While our pipeline can generate comprehensive scene and interaction data, methods for understanding such data remain unexplored. However, we demonstrated the data's utility through evaluation and plan to develop baseline models in future research. Second, the dataset requires validation in real-world environments, which we will address in future work through model performance evaluation in real-world settings.

7 CONCLUSION

In this work, we present a pipeline that generates human-scene interaction data in virtual environments using a user-centric scene graph structure. This approach reduces physical setup costs while providing comprehensive interaction annotations for MR tasks. Our pipeline successfully generated a dataset with detailed interaction annotations, creating diverse scene data by combining human-scene and hand-object interaction data through augmentation. The dataset has potential usage in interaction recognition, scene understanding, and human interaction simulation, enhancing MR systems' understanding of human-scene interactions. Future work will focus on developing baseline models and benchmarks, addressing hand-object penetration errors, and improving domain adaptation from synthetic to real-world environments.

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